Lecture 23

Types B and D tensor representations

Objectives. *We find simple $\mathfrak{so}(V)$-modules inside tensor powers of the vector module V, and the highest weights that these tensor powers miss.*

By the end of this lecture you will be able to:

- Prove $\operatorname{Sym}_0^k V \cong L(k\omega_1)$.
- Identify the exterior powers $\Lambda^k V$ with $k \le n$.
- Decide by computation which Hodge eigenspace carries $2\omega_n$.
- Decide which highest weights occur in tensor powers of V.

For $\mathfrak{sl}_m$, tensor powers of the defining module contain every finite-dimensional simple module (Lecture 18). We now ask the same question for the other classical semisimple algebras. The statement remains true for $\mathfrak{sp}_{2n}$ (Lecture 22). For $\mathfrak{so}_N$, tensor powers of the vector module miss the half-integral highest weights, so spin modules will be needed. This is a preview: Proposition 23.10 proves the statement for $\mathfrak{so}_N$, and Lecture 24 constructs the missing modules.

Throughout, $V = \mathbb{C}^N$ carries a nondegenerate symmetric bilinear form $(\cdot,\cdot)$. Either $N = 2n+1$ with $n \ge 2$, the type B_n, or $N = 2n$ with $n \ge 3$, the type D_n. All modules are finite-dimensional over $\mathbb{C}$.

23.1 Orthogonal root data

Recall from Lecture 10 that $\mathfrak{g} = \mathfrak{so}(V)$ consists of $X \in \mathfrak{gl}(V)$ with

$$(Xu, v) + (u, Xv) = 0 \quad \text{for all } u, v \in V.$$

Over $\mathbb{C}$ the form has an orthonormal basis $u_1, \dots, u_N$, in which $\mathfrak{g}$ consists of skew-symmetric matrices. Hence $\operatorname{tr} X = 0$ for $X \in \mathfrak{g}$.

Write $\mathrm{i} = \sqrt{-1}$ and set the isotropic basis

$$e_{\pm j} = \frac{u_j \pm \mathrm{i}\, u_{n+j}}{\sqrt{2}} \quad (1 \le j \le n), \qquad e_0 = u_N \ \text{ if } N \text{ is odd.}$$

Then

$$(e_j, e_{-k}) = \delta_{jk}, \qquad (e_j, e_k) = (e_{-j}, e_{-k}) = 0, \qquad (e_0, e_0) = 1,$$

and e_0 is orthogonal to every $e_{\pm j}$. Let E_{ab} denote matrix units in this basis. The diagonal elements

$$H(t) = \sum_{j=1}^{n} t_j \,(E_{jj} - E_{-j,-j})$$

form the Cartan subalgebra $\mathfrak{h}$ of Lecture 10. Put $\varepsilon_j(H(t)) = t_j$. The vector $e_{\pm j}$ has weight $\pm\varepsilon_j$, and e_0 has weight 0.

The positive root vectors from Lecture 10 are

$$\begin{aligned} X_{\varepsilon_j-\varepsilon_k} &= E_{jk} - E_{-k,-j} \quad (j<k),\\ X_{\varepsilon_j+\varepsilon_k} &= E_{j,-k} - E_{k,-j} \quad (j<k),\\ X_{\varepsilon_j} &= E_{j,0} - E_{0,-j} \quad (N \text{ odd}). \end{aligned}$$

The positive roots and the simple roots are

$$\begin{aligned} &R_+ = \{\varepsilon_j \pm \varepsilon_k : j<k\} \cup \{\varepsilon_j\} \text{ (only in } B_n),\\ &\alpha_j = \varepsilon_j - \varepsilon_{j+1}\ (j<n), \qquad \alpha_n = \varepsilon_n\ (B_n), \quad \alpha_n = \varepsilon_{n-1}+\varepsilon_n\ (D_n). \end{aligned}$$

The list of root vectors gives the raising property used below:

$$Xe_1 = 0, \qquad Xe_l \in \operatorname{span}(e_1,\dots,e_{l-1}) \quad (2 \le l \le n), \qquad \text{for all } X \in \mathfrak{n}_+. \tag{23.1}$$

Indeed $X_{\varepsilon_j-\varepsilon_k}$ with $j<k$ sends e_k to e_j, and the other positive root vectors kill every e_l with $l>0$. In D_n, only $X_{\varepsilon_j+\varepsilon_n}$ $(j<n)$ acts nontrivially on e_{-n}, so $Xe_{-n} \in \operatorname{span}(e_1,\dots,e_{n-1})$.

Use the classical coroots of Lecture 10 in these ε-coordinates. The fundamental weights are $\omega_k = \varepsilon_1+\cdots+\varepsilon_k$ for $k \le n-1$ in B_n and $k \le n-2$ in D_n, together with

$$\begin{aligned} B_n: \quad &\omega_n = \frac{1}{2}(\varepsilon_1+\cdots+\varepsilon_n),\\ D_n: \quad &\omega_{n-1} = \frac{1}{2}(\varepsilon_1+\cdots+\varepsilon_{n-1}-\varepsilon_n),\\ &\omega_n = \frac{1}{2}(\varepsilon_1+\cdots+\varepsilon_n). \end{aligned}$$

The conditions $\langle \rho, \alpha_j^\vee\rangle = 1$ for $1 \le j \le n$ give

$$\rho_B = (n-\frac{1}{2},\, n-\frac{3}{2},\, \dots,\, \frac{1}{2}), \qquad \rho_D = (n-1,\, n-2,\, \dots,\, 1,\, 0).$$

In both types $\rho_1 = (N-2)/2$. Integrality of a weight λ on the simple coroots forces all coordinates λ_j to be congruent modulo $\mathbb{Z}$, with $2\lambda_n \in \mathbb{Z}$. Hence

$$P = \mathbb{Z}^n \cup \big(\mathbb{Z}^n + (\frac{1}{2},\dots,\frac{1}{2})\big).$$

The simple coroots also give the dominance conditions. For $\lambda = \sum_j \lambda_j \varepsilon_j$,

$$\langle \lambda, \alpha_j^\vee\rangle = \lambda_j - \lambda_{j+1} \quad (j<n), \qquad \langle \lambda, \alpha_n^\vee\rangle = \begin{cases} 2\lambda_n & (B_n),\\ \lambda_{n-1}+\lambda_n & (D_n). \end{cases}$$

Hence a weight $\lambda \in P$ lies in P_+ exactly when

$$\lambda_1 \ge \cdots \ge \lambda_n \ge 0 \quad (B_n), \qquad \lambda_1 \ge \cdots \ge \lambda_{n-1} \ge |\lambda_n| \quad (D_n).$$

Its coordinates are then all integral or all half-integral. The last coordinate can be negative only in D_n.

Lemma 23.1 (Highest vectors generate simple submodules). *Let M be a finite-dimensional $\mathfrak{g}$-module and let $x \in M$ be a nonzero vector of weight λ with $\mathfrak{n}_+ x = 0$. Then $U(\mathfrak{g})x \cong L(\lambda)$.*

Proof. By PBW (Lecture 11),

$$M' = U(\mathfrak{g})x = U(\mathfrak{n}_-)U(\mathfrak{h})U(\mathfrak{n}_+)x = U(\mathfrak{n}_-)x.$$

The last step uses $\mathfrak{n}_+ x = 0$ and $Hx = \lambda(H)x$ for $H \in \mathfrak{h}$. Elements of positive degree in $U(\mathfrak{n}_-)$ lower weights strictly, so $M'_\lambda = \mathbb{C}x$.

The cyclic-module lemma of Lecture 13 applies to the finite-dimensional weight module M', generated by x with $M'_\lambda = \mathbb{C}x$. Hence $M' \cong L(\lambda)$. ∎

With $l = \lambda + \rho$ and $r = \rho$, the coroots of Lecture 10 turn the Weyl dimension formula of Lecture 17 into

$$\dim L(\lambda) = \prod_{j<k} \frac{l_j^2 - l_k^2}{r_j^2 - r_k^2} \cdot \beta, \qquad \beta = 1 \ (D_n), \quad \beta = \prod_j \frac{l_j}{r_j} \ (B_n). \tag{23.2}$$

23.2 Trace-free symmetric tensors

Identify $\operatorname{Sym} V$ with the polynomial algebra $\mathbb{C}[u_1, \dots, u_N]$. The algebra $\mathfrak{g}$ acts by derivations:

$$X(v_1 \cdots v_k) = \sum_{l=1}^{k} v_1 \cdots (Xv_l) \cdots v_k.$$

Definition 23.2 (Contraction and trace-free tensors)**.** For $k \geq 2$ the *contraction* $c \colon \operatorname{Sym}^k V \to \operatorname{Sym}^{k-2} V$ is

$$c(v_1 \cdots v_k) = \sum_{l<m} (v_l, v_m)\, v_1 \cdots \widehat{v_l} \cdots \widehat{v_m} \cdots v_k.$$

Put $c = 0$ on degrees 0 and 1. The *trace-free* tensors are $\operatorname{Sym}^k_0 V = \ker(c|_{\operatorname{Sym}^k V})$. Let $q = \sum_a u_a^2 \in \operatorname{Sym}^2 V$.

The map c is an intertwiner. In $c(X(v_1 \cdots v_k))$, the terms whose contracted pair avoids the factor Xv_l give $X\,c(v_1 \cdots v_k)$. The remaining terms sum to

$$\sum_{l<m} \big[(Xv_l, v_m) + (v_l, Xv_m) \big]\, v_1 \cdots \widehat{v_l} \cdots \widehat{v_m} \cdots v_k = 0.$$

The element q is invariant. Let $D(X)$ be the skew-symmetric matrix of X in the basis (u_a). Then

$$Xq = 2 \sum_a u_a\, Xu_a = 2 \sum_{a,b} D(X)_{ba}\, u_a u_b = 0,$$

since $D(X)_{ba} = -D(X)_{ab}$ and $u_a u_b = u_b u_a$.

In these coordinates $c = \frac{1}{2} \sum_a \partial_a^2$. For linear v, w we have $\sum_a \partial_a v\, \partial_a w = (v, w)$, and the Leibniz rule gives the displayed formula for c.

Lemma 23.3 (Harmonic decomposition · *[P]*)**.** *For $k \geq 2$ there is a decomposition into submodules*

$$\operatorname{Sym}^k V = \operatorname{Sym}^k_0 V \oplus q \operatorname{Sym}^{k-2} V, \qquad \dim \operatorname{Sym}^k_0 V = \binom{N+k-1}{k} - \binom{N+k-3}{k-2}.$$

Proof. Define a Hermitian form on $\operatorname{Sym} V$, conjugate-linear in the first argument, by its values on the monomials $u^\alpha = u_1^{\alpha_1}\cdots u_N^{\alpha_N}$:

$$\langle u^\alpha, u^\beta\rangle = \alpha!\,\delta_{\alpha\beta}, \qquad \alpha! = \alpha_1!\cdots\alpha_N!.$$

It is positive definite.

Both sides of $\langle u_a f, g\rangle = \langle f, \partial_a g\rangle$ vanish on the monomials $f = u^\alpha$ and $g = u^\beta$ unless $\beta = \alpha + \delta_a$, where δ_a is the ath unit multi-index. In that case both equal $\beta!$, since

$$\langle u^\alpha, \beta_a u^{\beta-\delta_a}\rangle = \beta_a\,\alpha! = \beta!.$$

Summing ∂_a^2 over a gives

$$\langle qf, g\rangle = 2\langle f, c(g)\rangle.$$

If $c(g) = 0$, then $g \perp q\operatorname{Sym}^{k-2} V$. Conversely, if $g \perp q\operatorname{Sym}^{k-2} V$, take $f = c(g)$ to get $c(g) = 0$. Hence $\operatorname{Sym}_0^k V$ is the orthogonal complement of $q\operatorname{Sym}^{k-2} V$.

Multiplication by q is injective, so the codimension is $\dim \operatorname{Sym}^{k-2} V$. The kernel of the intertwiner c is a submodule. Since q is invariant, $X(qf) = qXf$, so $q\operatorname{Sym}^{k-2} V$ is a submodule. ∎

Write the dimension in product form:

$$\begin{aligned}\binom{N+k-1}{k} - \binom{N+k-3}{k-2} &= \frac{(N+k-3)!}{k!\,(N-1)!}\big[(N+k-1)(N+k-2) - k(k-1)\big]\\ &= \frac{(N+k-3)!}{k!\,(N-1)!}\,(N-1)(2k+N-2)\\ &= \frac{(N+k-3)!\,(2k+N-2)}{k!\,(N-2)!}.\end{aligned}$$

Theorem 23.4 (Trace-free symmetric tensors · *[P]*). *For every $k \ge 0$, $\operatorname{Sym}_0^k V \cong L(k\omega_1)$ and*

$$\dim L(k\omega_1) = \frac{(k+N-3)!\,(2k+N-2)}{k!\,(N-2)!}.$$

Proof. The vector e_1^k has weight $k\varepsilon_1 = k\omega_1$. It is trace-free. For $k = 0, 1$, the contraction is zero by definition. For $k \ge 2$,

$$c(e_1^k) = \binom{k}{2}(e_1, e_1)e_1^{k-2} = 0.$$

The vector 1 is invariant. For $k \ge 1$, (23.1) gives

$$Xe_1^k = ke_1^{k-1}Xe_1 = 0 \qquad (X \in \mathfrak{n}_+).$$

By Lemma 23.1, $\operatorname{Sym}_0^k V$ contains a copy of $L(k\omega_1)$.

Now compute $\dim L(k\omega_1)$ by (23.2). Only $l_1 = k + r_1$ differs from r, so only the roots containing ε_1 contribute. For $2 \le j \le n$, both types have $r_1 - r_j = j - 1$ and $r_1 + r_j = N - 1 - j$. Hence

$$\dim L(k\omega_1) = \beta_k \prod_{j=2}^{n} \frac{k+j-1}{j-1} \cdot \prod_{j=2}^{n} \frac{k+N-1-j}{N-1-j},$$

where $\beta_k = 1$ for D_n and $\beta_k = (k + r_1)/r_1 = (2k+N-2)/(N-2)$ for B_n. Each factor has the form $(k+m)/m$, with

$$m \in \{1, \dots, n-1\} \text{ (first product)}, \qquad m \in \{N-1-n, \dots, N-3\} \text{ (second)}.$$

In B_n, $N-1-n=n$, so m runs once over $\{1,\dots,N-3\}$. In D_n, $N-1-n=n-1$, so $m=n-1$ occurs twice. The extra factor is $(k+n-1)/(n-1)=(2k+N-2)/(N-2)$. In both types

$$\dim L(k\omega_1)=\frac{(k+N-3)!}{k!\,(N-3)!}\cdot\frac{2k+N-2}{N-2},$$

which is the stated formula.

For $k\ge 2$ this equals $\dim \operatorname{Sym}^k_0 V$ by Lemma 23.3. For $k=0,1$ the spaces are $\mathbb{C}$ and V, of dimensions 1 and N. So the copy of $L(k\omega_1)$ is all of $\operatorname{Sym}^k_0 V$. ∎

23.3 Exterior powers

Write $e_{[k]}=e_1\wedge\cdots\wedge e_k$ in the exterior module $\Lambda^k V$ (Lecture 18), with the exterior action and antisymmetriser in Sheet 18.2.

We use the method of Lecture 21. A highest vector gives a simple submodule. We then compare its Weyl dimension with the dimension of the proposed ambient module. In the middle degree of D_n one highest vector is not enough. There we construct both highest-weight submodules and show that their dimensions add up to $\dim \Lambda^n V$.

Theorem 23.5 (Exterior powers · *[P]*). *(a) In B_n, for $1\le k\le n$, $\Lambda^k V\cong L(\varepsilon_1+\cdots+\varepsilon_k)$. This is $L(\omega_k)$ for $k<n$ and $L(2\omega_n)$ for $k=n$.*

(b) In D_n, for $1\le k\le n-1$, $\Lambda^k V\cong L(\varepsilon_1+\cdots+\varepsilon_k)$. This is $L(\omega_k)$ for $k\le n-2$ and $L(\omega_{n-1}+\omega_n)$ for $k=n-1$.

(c) In D_n, the highest vectors $e_{[n]}$ and $e_{[n-1]}\wedge e_{-n}$ generate

$$\Lambda^n V=L(2\omega_n)\oplus L(2\omega_{n-1}),\qquad \dim L(2\omega_n)=\dim L(2\omega_{n-1})=\frac{1}{2}\binom{2n}{n}.$$

Proof. Highest vectors. For $X\in\mathfrak{n}_+$,

$$Xe_{[k]}=\sum_{l=1}^{k} e_1\wedge\cdots\wedge Xe_l\wedge\cdots\wedge e_k.$$

By (23.1), Xe_l lies in the span of earlier factors, so each term vanishes. The same argument applies to $e_{[n-1]}\wedge e_{-n}$ in D_n, because $Xe_{-n}\in\operatorname{span}(e_1,\dots,e_{n-1})$. By Lemma 23.1, these vectors generate copies of $L(\varepsilon_1+\cdots+\varepsilon_k)$ and of $L(2\omega_{n-1})$.

Dimensions in D_n. For a finite set S of nonnegative numbers put $\Delta(S)=\prod_{a>b}(a^2-b^2)$, with $a,b\in S$. For $\lambda=\varepsilon_1+\cdots+\varepsilon_k$, the entries of $\lambda+\rho$ form $\{0,\dots,n\}\setminus\{m\}$, where $m=n-k$. For $2\omega_{n-1}$, $\lambda+\rho=(n,\dots,2,-1)$ has the same squares as for $k=n$. Removing m from $\{0,\dots,n\}$ divides Δ by $\prod_{b\ne m}|m^2-b^2|$, so (23.2) gives

$$\dim L(\lambda)=\frac{\prod_{b=0}^{n-1}(n^2-b^2)}{\prod_{0\le b\le n,\ b\ne m}|m^2-b^2|}.$$

The numerator is $n!\,(2n-1)!/(n-1)!=(2n)!/2$. For $m\ge 1$, the denominator is

$$\underbrace{m!\,(n-m)!}_{\prod|m-b|}\cdot\underbrace{\frac{(m+n)!}{2\,m!}}_{\prod(m+b)}=\frac{(n-m)!\,(n+m)!}{2},$$

so $\dim L(\lambda)=\binom{2n}{k}$. For $m=0$ the denominator is $(n!)^2$, and the dimension is $\frac{1}{2}\binom{2n}{n}$.

Dimensions in B_n. Double all coordinates. This leaves (23.2) unchanged. Then 2ρ has entries $\{1, 3, \dots, 2n-1\}$. Also $2(\lambda+\rho)$ has entries $\{1, 3, \dots, 2n+1\} \setminus \{p\}$, with $p = 2m+1$ and $m = n-k$. As before,

$$\dim L(\lambda) = \frac{(2n+1)\prod_{b\le 2n-1,\ b \text{ odd}}\big((2n+1)^2 - b^2\big)}{p\prod_{b\le 2n+1,\ b \text{ odd},\ b\neq p}|p^2-b^2|}.$$

Here b runs over positive odd numbers. With $b = 2j-1$ in the numerator,

$$\begin{aligned}(2n+1)^2 - (2j-1)^2 &= 4(n+1-j)(n+j),\\ \prod_{b\neq p}|p-b| &= 2^n\, m!\,(n-m)!,\\ p\prod_{b\neq p}(p+b) &= 2^n\,(n+m+1)!/m!.\end{aligned}$$

Hence

$$\dim L(\lambda) = \frac{4^n(2n+1)!}{4^n(n-m)!\,(n+m+1)!} = \binom{2n+1}{k}.$$

Conclusion. In (a) and (b) the generated copy has dimension $\binom{N}{k} = \dim \Lambda^k V$, so it is everything. In (c) the two copies are nonisomorphic irreducibles, so they meet in zero. Their dimensions add to $\binom{2n}{n} = \dim \Lambda^n V$. ∎

23.4 The middle degree and the Hodge operator

Theorem 23.5 splits $\Lambda^n V$ in D_n by choosing highest vectors. The Hodge operator gives the same splitting by an eigenvalue, without choosing a highest vector. It also identifies $\Lambda^{N-k}V$ with $\Lambda^k V$, so exterior powers above the middle degree give no new modules. Lecture 25 returns to this splitting of $\Lambda^n V$ when it decomposes the square of a spin module.

On $\Lambda^k V$ use the form

$$(a_1 \wedge \cdots \wedge a_k,\ b_1 \wedge \cdots \wedge b_k) = \det\big((a_l, b_m)\big).$$

It is invariant. For $\alpha = a_1 \wedge \cdots \wedge a_k$ and $\beta = b_1 \wedge \cdots \wedge b_k$, let C_{lm} be the (l,m) cofactor of $\big((a_l, b_m)\big)$. Expanding along rows, then along columns, and using invariance on V gives

$$(X\alpha, \beta) = \sum_{l,m}(Xa_l, b_m)\, C_{lm} = -(\alpha, X\beta).$$

The wedges $u_I = u_{i_1} \wedge \cdots \wedge u_{i_k}$ with $i_1 < \cdots < i_k$ form an orthonormal basis of $\Lambda^k V$.

Definition 23.6 (Hodge operator). Fix $\mathrm{vol} \in \Lambda^N V$ with $(\mathrm{vol}, \mathrm{vol}) = 1$. The *Hodge operator* $\star\colon \Lambda^k V \to \Lambda^{N-k} V$ is defined by

$$\alpha \wedge \star\beta = (\alpha, \beta)\,\mathrm{vol} \qquad \text{for all } \alpha \in \Lambda^k V.$$

The wedge pairing $\Lambda^k \times \Lambda^{N-k} \to \Lambda^N$ is perfect, so $\star$ exists and is unique. It sends u_I to $\pm u_{I^c}$, so it is bijective. It is also an intertwiner. Since $\operatorname{tr} X = 0$, we have $X\mathrm{vol} = 0$, and

$$\begin{aligned}\alpha \wedge X(\star\beta) &= X(\alpha \wedge \star\beta) - X\alpha \wedge \star\beta\\ &= -(X\alpha, \beta)\,\mathrm{vol} = (\alpha, X\beta)\,\mathrm{vol} = \alpha \wedge \star(X\beta).\end{aligned}$$

Thus $X(\star\beta) = \star(X\beta)$, and $\Lambda^{N-k}V \cong \Lambda^k V$ for every k.

Proposition 23.7 (Hodge eigenvalues · *[P]*). *Let $N = 2n$ and $\mathrm{vol} = u_1 \wedge u_{n+1} \wedge u_2 \wedge u_{n+2} \wedge \cdots \wedge u_n \wedge u_{2n}$. Put $c_n = \mathrm{i}^n(-1)^{n(n+1)/2}$. Then*

$$\star e_{[n]} = c_n\, e_{[n]}, \qquad \star\big(e_{[n-1]} \wedge e_{-n}\big) = -c_n\, e_{[n-1]} \wedge e_{-n}.$$

Hence $\star$ acts by c_n on $L(2\omega_n)$ and by $-c_n$ on $L(2\omega_{n-1})$.

Proof. The operator $\star$ commutes with $\mathfrak{h}$. The weight $(1,\dots,1)$ occurs in $\Lambda^n V$ only for $e_{[n]}$. Hence $\star e_{[n]} = \gamma\, e_{[n]}$ for some γ. Pair with $\alpha = e_{-1} \wedge \cdots \wedge e_{-n}$. Then $(\alpha, e_{[n]}) = \det I = 1$, so

$$\gamma\, \alpha \wedge e_{[n]} = \mathrm{vol}.$$

From $e_j \wedge e_{-j} = -\mathrm{i}\, u_j \wedge u_{n+j}$ we get

$$\mathrm{vol} = \mathrm{i}^n\, e_1 \wedge e_{-1} \wedge \cdots \wedge e_n \wedge e_{-n}.$$

Move e_1 to the front past n factors. Then move e_2 past $n-1$ factors, and so on. This gives

$$\alpha \wedge e_{[n]} = (-1)^{n(n+1)/2}\, e_1 \wedge e_{-1} \wedge \cdots \wedge e_n \wedge e_{-n}.$$

Comparing the two expressions gives $\gamma = c_n$.

Likewise the weight $(1,\dots,1,-1)$ occurs in $\Lambda^n V$ only for $e_{[n-1]} \wedge e_{-n}$, so $\star$ multiplies it by some γ'. Pair with $\alpha' = e_{-1} \wedge \cdots \wedge e_{-(n-1)} \wedge e_n$, for which $(\alpha', e_{[n-1]} \wedge e_{-n}) = 1$. The product $\alpha' \wedge e_{[n-1]} \wedge e_{-n}$ is $\alpha \wedge e_{[n]}$ with e_n and e_{-n} exchanged, so it has the opposite sign. Hence $\gamma' = -c_n$.

Since $\star$ commutes with $\mathfrak{g}$, it acts by c_n and $-c_n$ on the submodules generated by these two vectors. These are the two summands of $\Lambda^n V$ in Theorem 23.5. ∎

For odd n the eigenvalues are $\pm\mathrm{i}$, and $\mathrm{i}\star$ is an involution. Replacing vol by $-$vol negates $\star$. With vol fixed, exchanging e_n and e_{-n} replaces ε_n by $-\varepsilon_n$. It exchanges α_{n-1} and α_n, and hence the labels $2\omega_{n-1}$ and $2\omega_n$. An eigenvalue names a highest weight only after both choices are fixed.

Theorem 23.8 (Orthogonal harmonic Schur modules · *[C]*). *Let λ be a partition of d with column lengths $\lambda'_1 \ge \lambda'_2 \ge \dots$. Put $\mathbb{S}_{[\lambda]}V = V^{[d]} \cap \mathbb{S}_\lambda V$, where $V^{[d]}$ is the common kernel of all pair contractions on $V^{\otimes d}$ and $\mathbb{S}_\lambda V$ is the image of a Young symmetriser.*

(i) $\mathbb{S}_{[\lambda]}V \neq 0$ *exactly when* $\lambda'_1 + \lambda'_2 \le N$.

(ii) If $\lambda'_1 \le n$ *in* B_n, *or* $\lambda'_1 \le n-1$ *in* D_n, *then* $\mathbb{S}_{[\lambda]}V \cong L(\lambda_1,\dots,\lambda_n)$.

(iii) If $\lambda'_1 = n$ *in* D_n, *then* $\mathbb{S}_{[\lambda]}V$ *splits as* $L(\lambda_1,\dots,\lambda_n) \oplus L(\lambda_1,\dots,\lambda_{n-1},-\lambda_n)$.

Source: Fulton–Harris, §19.5, for $\mathrm{SO}_N(\mathbb{C})$. The passage to $\mathfrak{g}$ is not proved here. No argument below uses it. The theorem completes the tensor picture, in parallel with the symplectic Weyl construction of Lecture 22.

23.5 The vector square of $\mathfrak{so}_8$ and the missing weights

Example 23.9 ($8_v \otimes 8_v$). Let $N = 8$ and $n = 4$, so $V = 8_v = L(\omega_1)$. The flip is an intertwiner (Lecture 14), so

$$V \otimes V = \mathrm{Sym}^2 V \oplus \Lambda^2 V = \mathrm{Sym}^2_0 V \oplus \mathbb{C}q \oplus \Lambda^2 V.$$

By Theorem 23.4 with $k = 2$, $\mathrm{Sym}^2_0 V \cong L(2\omega_1)$ has dimension

$$\dim L(2\omega_1) = \frac{7! \cdot 10}{2!\,6!} = \frac{7 \cdot 10}{2} = 35.$$

By Theorem 23.5 with $k = 2 \le n - 2$,

$$\Lambda^2 V \cong L(\omega_2), \qquad \dim \Lambda^2 V = \binom{8}{2} = \frac{8 \cdot 7}{2} = 28.$$

Put $35_v = L(2\omega_1)$. Hence

$$8_v \otimes 8_v = 1 \oplus 28 \oplus 35_v, \qquad 64 = 1 + 28 + 35.$$

Proposition 23.10 (Highest weights in $\mathbb{Z}^n$ · *[P]*). *For $\lambda \in P_+$ the following are equivalent:*

(i) $L(\lambda)$ is isomorphic to a submodule of $V^{\otimes k}$ for some $k \ge 0$, where $V^{\otimes 0} = \mathbb{C}$;

(ii) every weight of $L(\lambda)$ lies in $\mathbb{Z}^n$;

(iii) $\lambda \in \mathbb{Z}^n$.

Proof. (i)⇒(ii): each weight of $V^{\otimes k}$ is a sum of k weights in $\{\pm\varepsilon_j, 0\}$. (ii)⇒(iii): λ is a weight.

(iii)⇒(i): put $f_j = \varepsilon_1 + \cdots + \varepsilon_j$, $\lambda_{n+1} = 0$ and, in D_n, $f_n' = f_{n-1} - \varepsilon_n$. The case $\lambda_n < 0$ occurs only in D_n. Then

$$\lambda = \sum_{j=1}^{n} (\lambda_j - \lambda_{j+1}) f_j \qquad (\lambda_n \ge 0),$$

$$\lambda = \sum_{j=1}^{n-2} (\lambda_j - \lambda_{j+1}) f_j + (\lambda_{n-1} + \lambda_n) f_{n-1} - \lambda_n f_n' \qquad (\lambda_n < 0).$$

The dominance inequalities of Section 23.1 make the coefficients nonnegative integers. By Theorem 23.5, each f_j is a highest weight in $\Lambda^j V$. In D_n, f_n' is a highest weight in $\Lambda^n V$. The antisymmetriser in Sheet 18.2 embeds these spaces in tensor powers of V. A tensor product of such vectors is nonzero, has weight λ, and is killed by $\mathfrak{n}_+$, which acts by derivations. Lemma 23.1 gives $L(\lambda) \subseteq V^{\otimes k}$, where k is the sum of the exterior degrees used. ∎

Figure 23.1 shows the proposition for B_2, where $\omega_1 = (1, 0)$ and $\omega_2 = (\frac{1}{2}, \frac{1}{2})$.

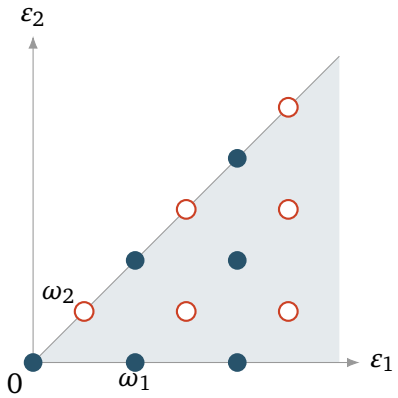


Figure 23.1: Dominant weights $x\varepsilon_1 + y\varepsilon_2$ of B_2 with $x \le \frac{5}{2}$, in the chamber $x \ge y \ge 0$. Filled points lie in $\mathbb{Z}^2$ and open points in $\mathbb{Z}^2 + (\frac{1}{2}, \frac{1}{2})$; each point is the highest weight of a different module, and only the filled points label modules inside tensor powers of V.

For $\mathfrak{so}_8$, the spin weights $\omega_3 = \frac{1}{2}(1, 1, 1, -1)$ and $\omega_4 = \frac{1}{2}(1, 1, 1, 1)$ do not lie in $\mathbb{Z}^4$. By Proposition 23.10, $L(\omega_3)$ and $L(\omega_4)$ lie in no tensor power of 8_v. Lecture 24 constructs them from a Clifford algebra.

Self-checks

(1) Give the highest weight of $\Lambda^3\mathbb{C}^7$ for $\mathfrak{so}_7$ in Dynkin labels, and its dimension.

(2) Does $L(\omega_1+\omega_4)$ for $\mathfrak{so}_8$ lie in a tensor power of 8_v?

(3) Replace vol by $-$vol for $\mathfrak{so}_8$. On which half of $\Lambda^4 V$ does $\star$ now act by $+1$?

Answers. (1) $\varepsilon_1+\varepsilon_2+\varepsilon_3 = 2\omega_3$, labels $(0,0,2)$, dimension $\binom{7}{3} = 35$. (2) No: $\omega_1+\omega_4 = (\frac{3}{2},\frac{1}{2},\frac{1}{2},\frac{1}{2}) \notin \mathbb{Z}^4$. (3) On $L(2\omega_3)$, since $c_4 = 1$ and $\star$ changes sign.

Summary

Trace-free symmetric tensors give $L(k\omega_1)$. Exterior powers give $L(\varepsilon_1+\cdots+\varepsilon_k)$ for $k \le n$ in B_n and for $k \le n-1$ in D_n. In D_n, the middle exterior power splits into Hodge eigenspaces. Tensor powers reach exactly the dominant weights in $\mathbb{Z}^n$. Spin modules follow in Lecture 24.

Reading

W. Fulton and J. Harris, *Representation Theory: A First Course*, §§18–19. R. Carter, *Lie Algebras of Finite and Affine Type*, §§13.3–13.4. R. Goodman and N. R. Wallach, *Symmetry, Representations, and Invariants*, Chapter 10.